ZERO-ONE HACK 2026 · INFINEON · INDUSTRIAL AI

Process Logic, Learned.

A hybrid neural–symbolic pipeline that learns the hidden grammar of semiconductor manufacturing — and proves it.

Next-step | Completion | Anomaly detection | + hidden 4th-family OOD

Team TBD

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THE PROBLEM

A chip is a 150-step sentence where order is physics.

- Every wafer runs an ordered route: **clean → oxidize → pattern → etch → implant → … → test → ship** (~115–150 steps, 3 families).
- Order encodes real constraints: **no etch before a mask, no deposit on a dirty surface, no ship before sort-test** — 10 long-range rules.
- The judges' real question: does the model learn the process **logic**, or just memorize patterns?
- And it's graded on a **hidden 4th family** never seen in training.

THE INSIGHT

Treat a recipe like a sentence. One step = one word.

TOKENIZER

A custom atomic-step tokenizer — each whole step is one token (vocab 210). No sub-word fragmentation; recipes read and write like language.

The Writer · DECODER

A from-scratch Llama-style causal Transformer that generates — next step & completion (Tasks 1–2). Attention does long-range lookups an LSTM can't.

The Judge · ENCODER

A from-scratch DeBERTa-v2 encoder that classifies validity + which rule broke (Task 3), trained with contrastive hard-negative twins.

Two specialized Transformers, trained from scratch — no pretrained model, full control of the vocabulary, interpretable representations.

OUR APPROACH · THE HYBRID

The model proposes. Physics disposes.

NEURAL — PLAUSIBILITY

Decoder + Encoder

Learn the grammar from data. Rank likely next steps, generate fluent completions, and output a real probability of validity.

SYMBOLIC — CORRECTNESS

Physics Refinery

A deterministic rule engine (≡ the official validator). Vetoes illegal steps, repairs dead-ends, certifies the verdict.

OUTPUT

Best of both

Guaranteed-valid routes that are also likely — “the model decided, the rules certify it.”

Neural alone can be confidently wrong; rules alone can't say what's likely. The hybrid covers both blind spots.

HOW WE TACKLED THE 3 TASKS

One hybrid, wired three ways.

1 Next-step

Decoder's top-5 candidates, then the Refinery floats the legal ones to the top. Cannot hurt a good model; rescues a bad rank.

2 Completion

Decoder proposes; rule-vetoed beam search + repair (beam 5, branch 8). Every emitted suffix is provably valid.

3 Anomaly

Contrastive encoder learns WHY a route is invalid; rule engine gives the verdict, the model gives the graded score (real ROC-AUC).

Generative tasks use the writer; the discriminative task uses the judge — each with a guarantee the learned model can't make alone.

RESULTS · OFFICIAL SCORER, HELD-OUT SPLIT

It works — and we can prove it learned.

Task	Headline	Also
1 · Next-step	Top-1 0.937 · Top-5 1.000	MRR 0.97 (hybrid)
2 · Completion	Block-level 0.94 · 100% rule-valid	token 0.71 · exact 0.28
3 · Anomaly	F1 1.000 · rule-attr 0.972	encoder alone AUC 0.49 (honest)
Understanding	0.998 next-operation acc	learns function, not just names

learns **function**, not just names: predicts the right **operation 99.8% of the time** · completions are 100% rule-valid

Honest provenance: real numbers from the organizer's official scorer on a held-out self-eval split, from the completed ProcSeq Leonardo run (base, 16k steps). The learned anomaly encoder alone is ~chance (AUC 0.49) — the physics hybrid carries Task 3. Organizer hidden-test and 4th-family OOD are pending.

RESULTS · DID IT ACTUALLY LEARN?

Not just a score — evidence it learned the logic.

Learns the operation

0.998 next-OPERATION accuracy — it predicts the right kind of step (deposit / etch / clean), not just a memorized name. That's process logic, not pattern recall.

Always valid

100% of completions satisfy all 10 rules. Physics-vetoed beam search + repair means 0 illegal outputs — a guarantee a neural model alone can't give.

Honest on the hard task

The learned anomaly encoder alone is ~chance (AUC 0.49); the physics hybrid supplies the exact verdict (F1 1.0, rule-attr 0.97). We show both, not just the flattering one.

Every number is held-out and official-scorer; the organizer hidden test and 4th-family OOD are marked pending — we report what's measured.

WHAT WE'D DO NEXT

Clear next steps, already scoped.

- **Scale the curve** — two larger model sizes / more steps to chart accuracy vs. compute.
- **Ontology input channel** — feed each step's physical category into the encoder so it reads an unseen 4th family.
- **The honest OOD curve** — run the novel-vocabulary stress test on the trained models (rename fraction 0 → 1).
- **Calibrate + per-family** — calibrate the hybrid anomaly score and add a per-family results breakdown.

ProcSeq — process logic, learned, guaranteed, and honestly benchmarked. **Thank you.**